

An introduction to statistical inference

Francisco Rodríguez-Sánchez

<https://frodriguezsanchez.net>

Why statistics?

To answer questions like...

- what's the probability that something occurs?

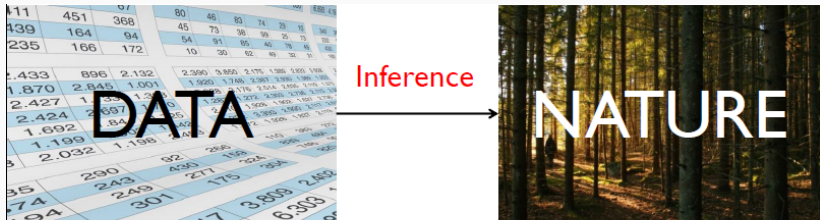
To answer questions like...

- what's the probability that something occurs?
- does X influence Y? How much?

To answer questions like...

- what's the probability that something occurs?
- does X influence Y? How much?
- can we predict Y knowing X, Z... How well?

To ensure correct inferences



Bolker et al 2009 TREE:

'311 out of 537 GLMM analyses (58%) used these tools inappropriately'

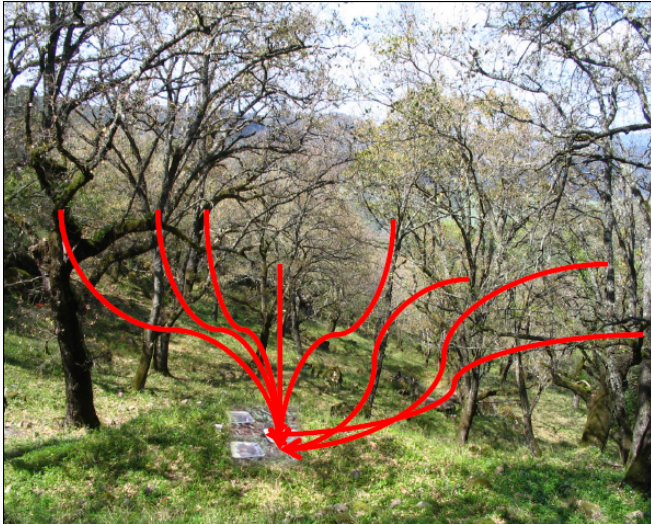
To get answers to tough problems

How many seeds do trees produce?



A. Torrenegra

Inferring tree fecundity



Course goals

- Understand statistical inference

Course goals

- Understand statistical inference
- Avoid misconceptions

Course goals

- **Understand** statistical inference
- Avoid **misconceptions**
- Promote **good practices**

*The purpose of models is not to fit data
but to sharpen thinking*

Sam Karlin

- Descriptive statistics

Topics

- Descriptive statistics
- Graphics

Topics

- Descriptive statistics
- Graphics
- Sampling

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- Descriptive statistics
- Graphics
- Sampling
- Experimental design

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- Hypothesis testing

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- Bayesian inference

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- Linear models & GLMs

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- Descriptive statistics
- Graphics
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- Experimental design
- Hypothesis testing
- Bayesian inference
- Linear models & GLMs
- Model selection

Descriptive statistics

Measure trunk diameter of 30 trees in your neighbourhood



Read data

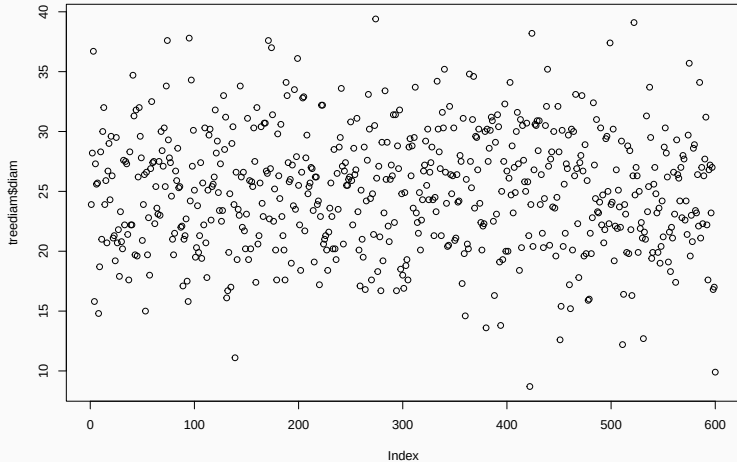
```
treediam <- read.csv("treediam.csv")
```

```
summary(treediam)
```

site	tree	diam
Min. : 1.00	Min. : 1.0	Min. : 8.70
1st Qu.: 5.75	1st Qu.: 8.0	1st Qu.:21.40
Median :10.50	Median :15.5	Median :25.25
Mean :10.50	Mean :15.5	Mean :25.04
3rd Qu.:15.25	3rd Qu.:23.0	3rd Qu.:28.40
Max. :20.00	Max. :30.0	Max. :39.40

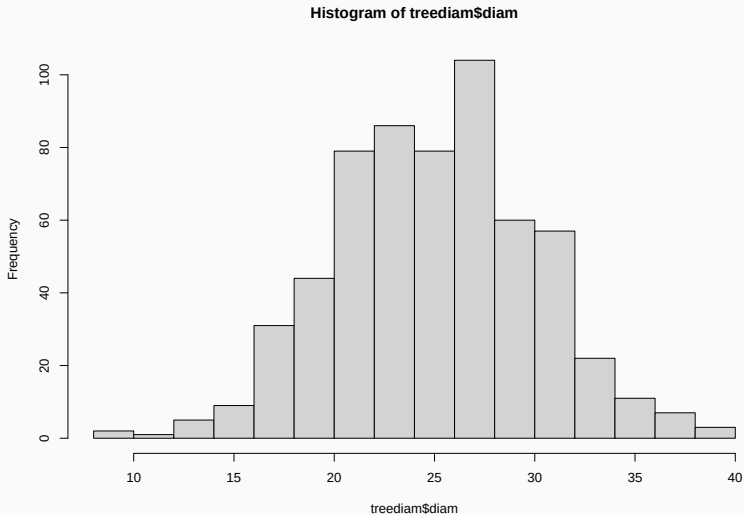
Visualisation of tree diameters

```
plot(treediam$diam)
```



Visualisation of tree diameters

```
hist(treediam$diam)
```



How well do these values
represent actual tree diameters
in your neighbourhood?

<https://pollev.com/franciscorod726>

- At what height did you measure?

- At what height did you measure?
- Did you include bark?

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- Did you measure with tape, caliper, by eye?

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 - (trees may get thinner w/ high evapotranspiration)

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- Where did you measure?

- At what height did you measure?
- Did you include bark?
- Did you measure with tape, caliper, by eye?
- When did you measure: dawn, midday, night?
 - (trees may get thinner w/ high evapotranspiration)
- Where did you measure?
 - (differences among streets, species, etc)



TRUTH



TRUTH



TRUTH

<https://visualizevalue.com/>

Data are hardly ever objective.

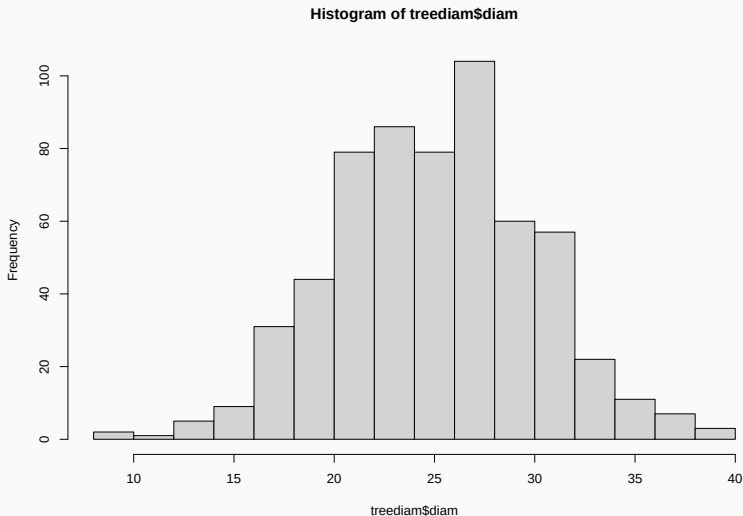
We decide **what to measure, when, where, and how.**

Always consider:

How well do data reflect what we are trying to measure?

Describing your data

How would you describe this distribution?

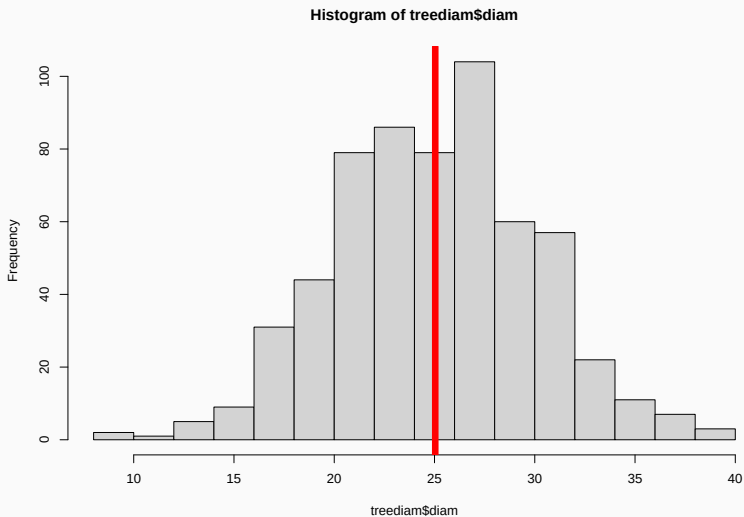


(Discuss with your partner)

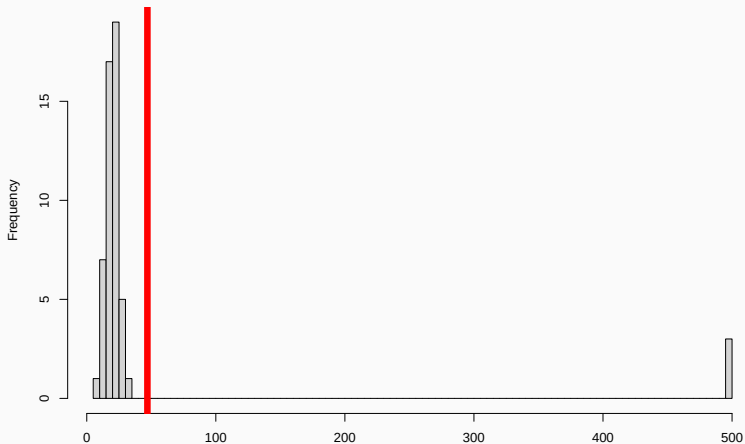
Location / Central tendency

Mean / Average

$$\text{mean} = \frac{d_1 + d_2 + d_3}{n}$$

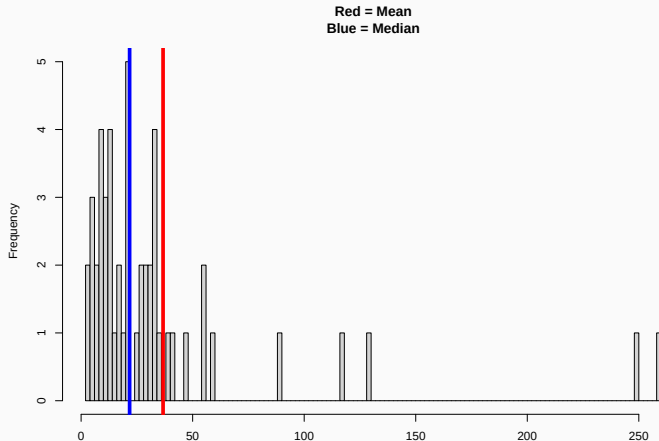


Mean is sensitive to skew/outliers



Median

50% percentile. Leaves half of the data values on each side

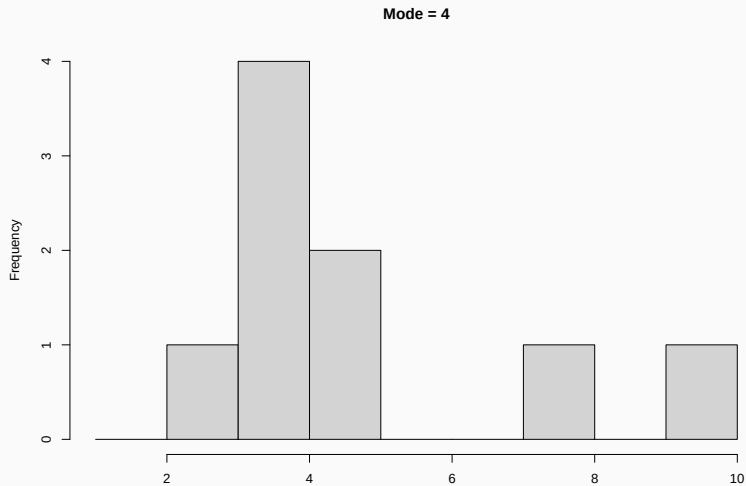


Median of $c(2, 4, 6, 8, 10) = 6$

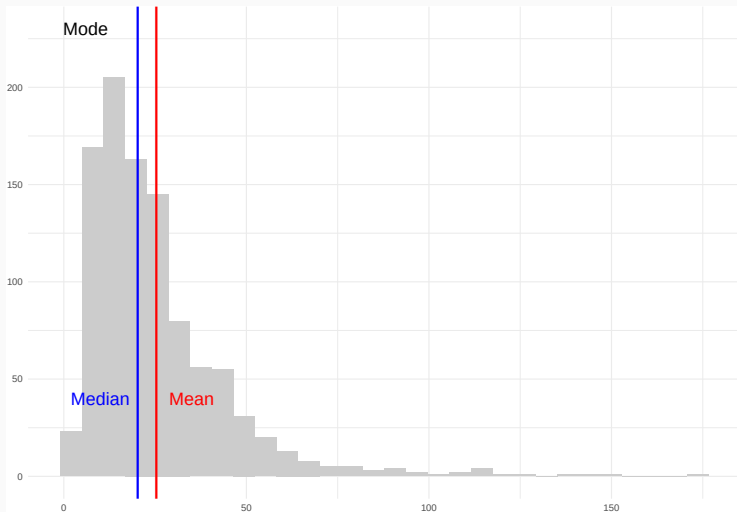
Median of $c(2, 4, 6, 8) = (4 + 6) / 2 = 5$

Mode

Most frequent value

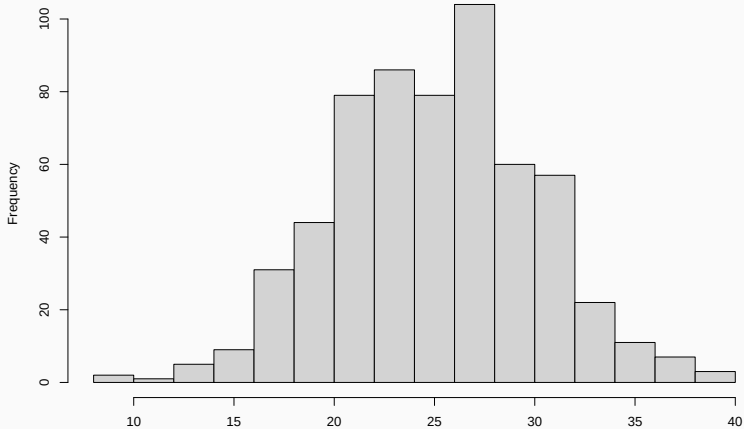


Describing the location / central tendency



Describing Variation / Spread

Minimum, Maximum, Range



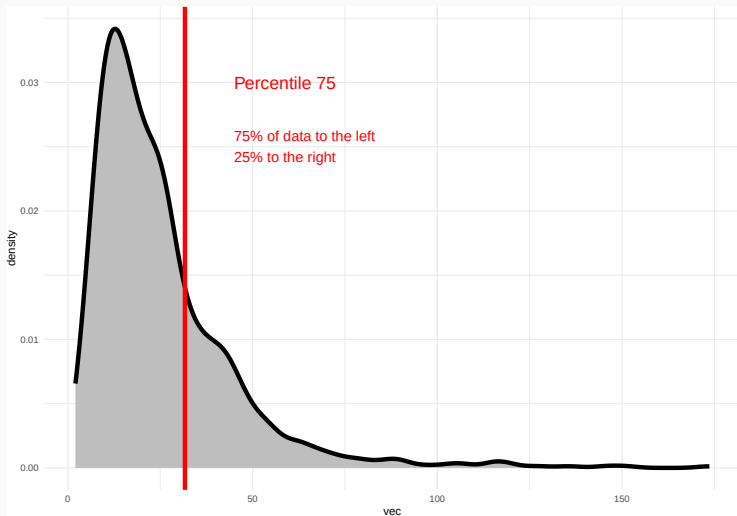
Minimum = 9.2

Maximum = 41.9

Range = 9.2, 41.9

Quantiles

Quartiles, Percentiles...

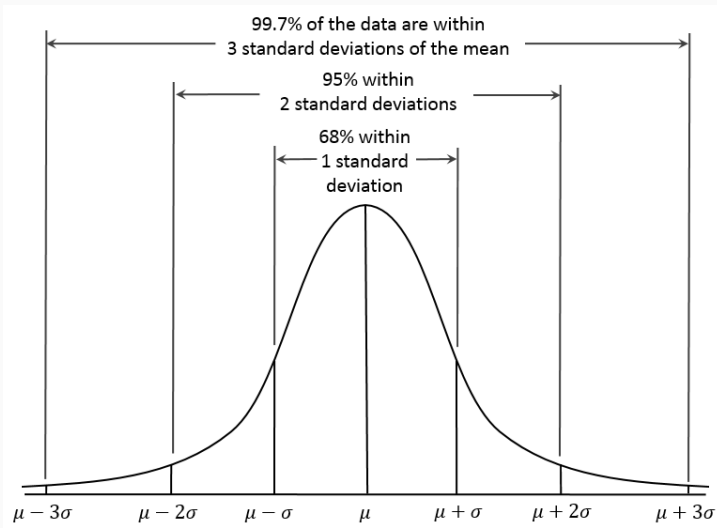


Standard Deviation

Average distance between data points and the mean

$$SD = \sqrt{\frac{\sum (x - \mu)^2}{n - 1}}$$

In a Normal distribution

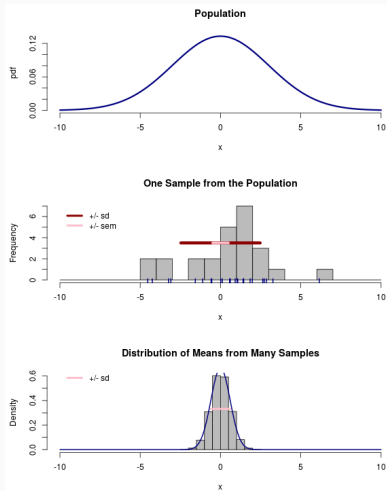


Standard Error of the Mean

$$SEM = \frac{SD}{\sqrt{n}}$$

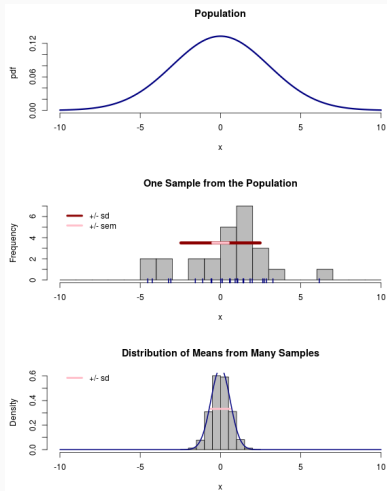
Estimates uncertainty (spread) of the parameter 'mean'

Relationship between SD and SEM



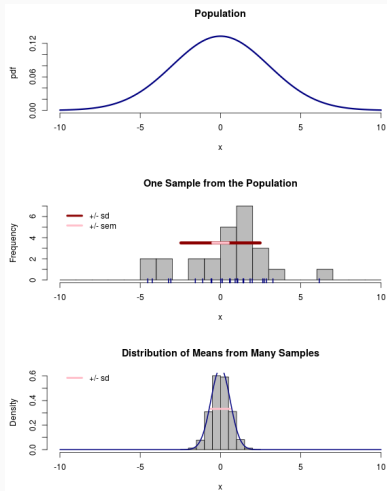
- https://gallery.shinyapps.io/sampling_and_stderr/

Relationship between SD and SEM



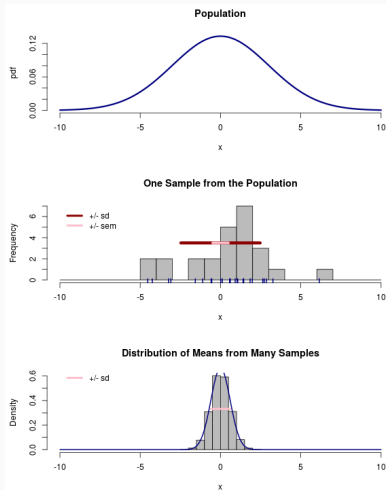
- https://gallery.shinyapps.io/sampling_and_stderr/
- SD quantifies scatter in population

Relationship between SD and SEM



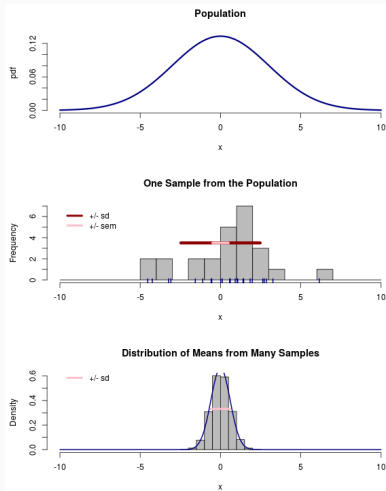
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Relationship between SD and SEM



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- $SEM = SD / \sqrt{n}$

Relationship between SD and SEM

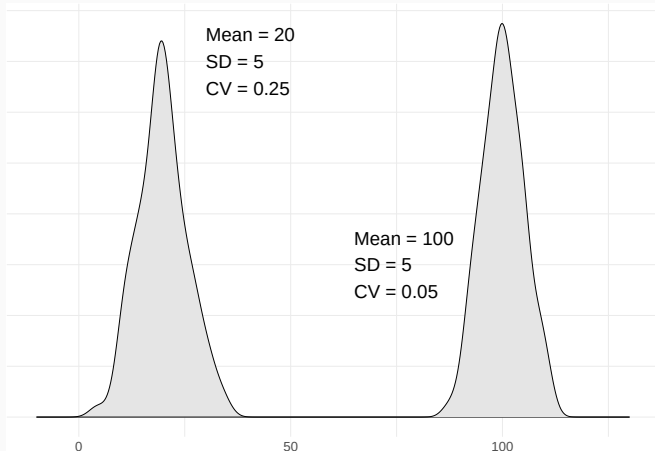


- https://gallery.shinyapps.io/sampling_and_stderr/
- SD quantifies scatter in population
- SEM quantifies uncertainty in parameter estimate (population mean)
- $SEM = SD / \sqrt{n}$
- SEM decreases with sample size (mean better known), SD does not.

Coefficient of Variation

Facilitates comparing spread of distributions with different means

$$CV = \frac{SD}{mean}$$

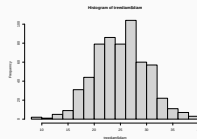


Summarise a distribution

Central tendency / location

- mean (average)

Variation / Spread

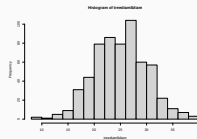


Summarise a distribution

Central tendency / location

- mean (average)
- median (50% percentile)

Variation / Spread

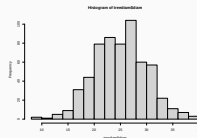


Summarise a distribution

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Variation / Spread



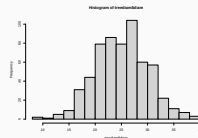
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Variation / Spread

- min, max, range



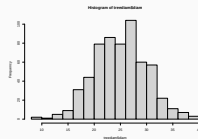
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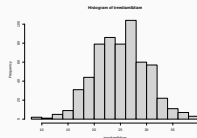
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- standard deviation



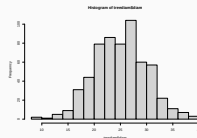
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- standard error of the mean



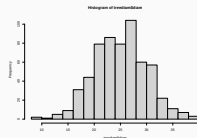
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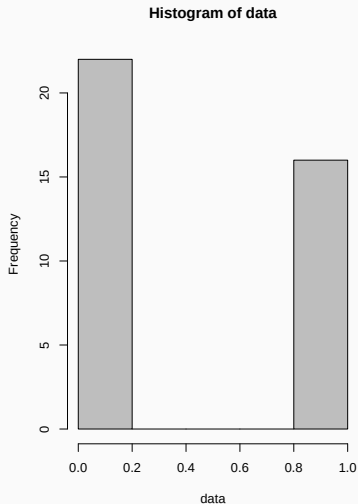
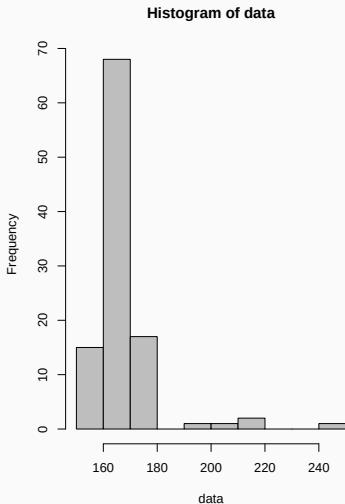
Variation / Spread

- min, max, range
- quantiles (quartiles, percentiles...)
- standard deviation
- standard error of the mean
- coefficient of variation



What statistical descriptors are best? (and why)

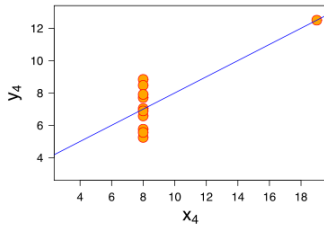
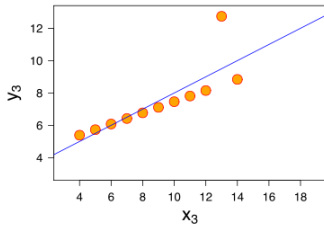
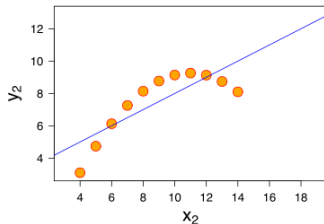
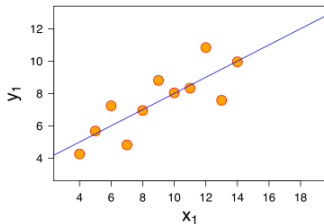
<https://pollev.com/franciscorod726>



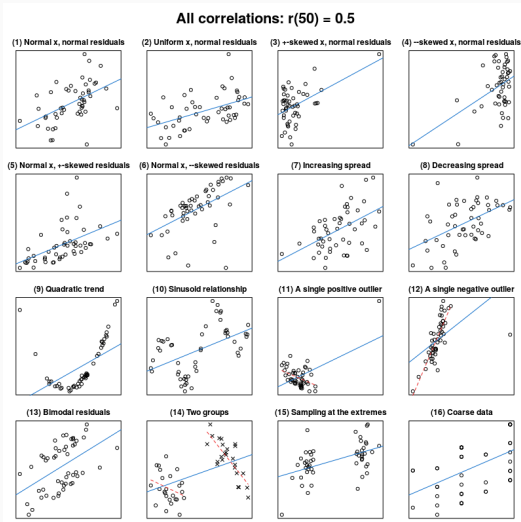
The importance of visualisation

Visualisation of data & models is
key

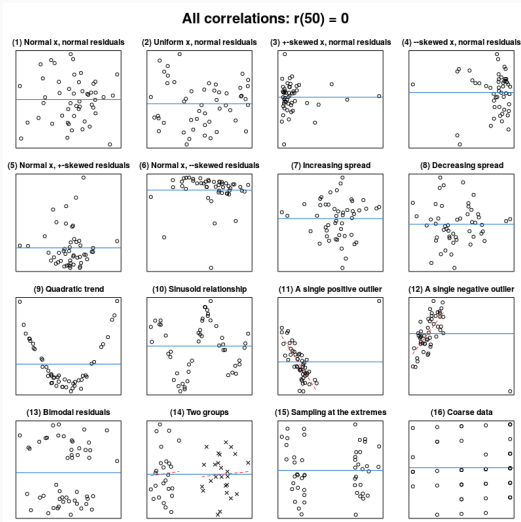
Always plot data and models



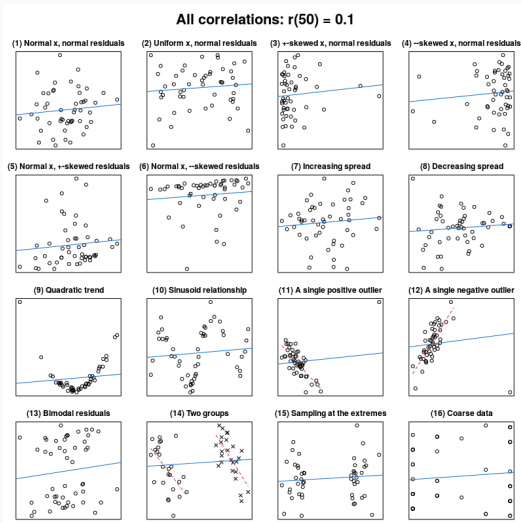
Don't use statistics blindly: *Visualise*



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Don't use statistics blindly: *Visualise*



<https://janhove.github.io/teaching/2016/11/21/what-correlations-look-like>

Plot. Check models. Plot. Check assumptions. Plot.

Lavine 2014 Ecology

Sampling, confidence intervals, likelihood and Bayesian inference

Inference: from samples to population

We rarely measure the whole **population**, but take **samples**.

Then we make inferences from sample to population.



If we sample 30 trees in our neighbourhood...

Can we extrapolate results to

- whole neighbourhood?
- whole city?
- whole country?
- the world?

What's the **suitable population** to make inferences given this sample?

Understanding confidence intervals

- Summarise **uncertainty** in parameter estimates.

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- It's a frequentist, long-run property.
- To read more: [Morey et al \(2015\)](#), [Rohrer 2024](#)

What happens if we increase sample size?

<https://rpsychologist.com/d3/CI/>

- CI width *decreases*...

What happens if we increase sample size?

<https://rpsychologist.com/d3/CI/>

- CI width *decreases*...
- but still 5% of CIs will NOT contain true mean!

If the 95% CI of X is (120, 150)...

- There is a 95% probability that X lies between 120 and 150

<https://pollev.com/franciscorod726>

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- If we repeated the experiment, 95% of the time X would fall between 120 and 150

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- If we repeated the experiment, 95% of the time X would fall between 120 and 150
- If we repeated the experiment, 95% of the CIs would contain the true value of X
- The probability that X is greater than 0 is at least 95%

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- We can be 95% confident that X lies between 120 and 150
- If we repeated the experiment, 95% of the time X would fall between 120 and 150
- If we repeated the experiment, 95% of the CIs would contain the true value of X
- The probability that X is greater than 0 is at least 95%
- The probability that X equals 0 is smaller than 5%

<https://pollev.com/franciscorod726>

Bayesian credible intervals

- Bayesian **credible** intervals do give the probability that true parameter value is contained within them.

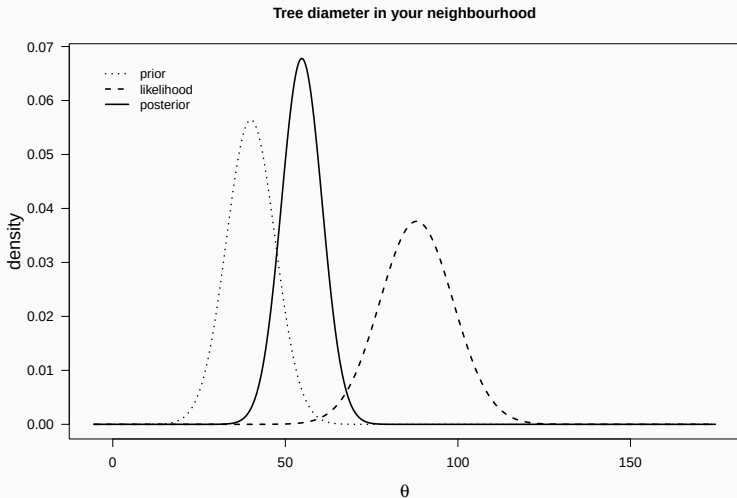
Bayesian credible intervals

- Bayesian **credible** intervals do give the probability that true parameter value is contained within them.
- Frequentist CIs and Bayesian credible intervals can be similar, but not always.

Bayesian inference: prior, posterior, and likelihood

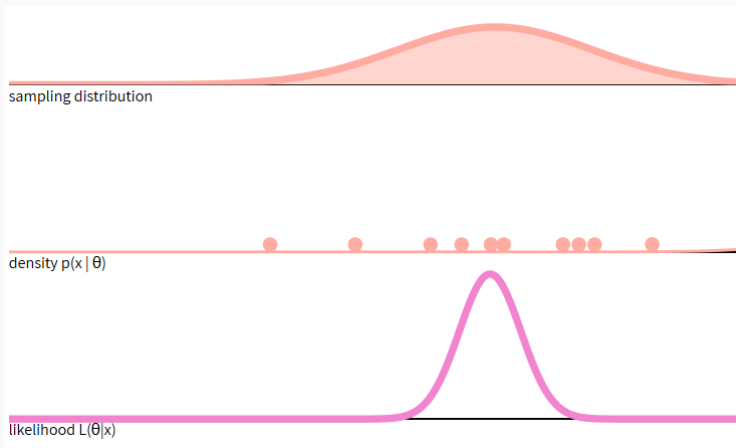
$$P(\text{Unknown}|\text{Data}) \propto P(\text{Data}|\text{Unknown}) \times P(\text{Unknown})$$

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$



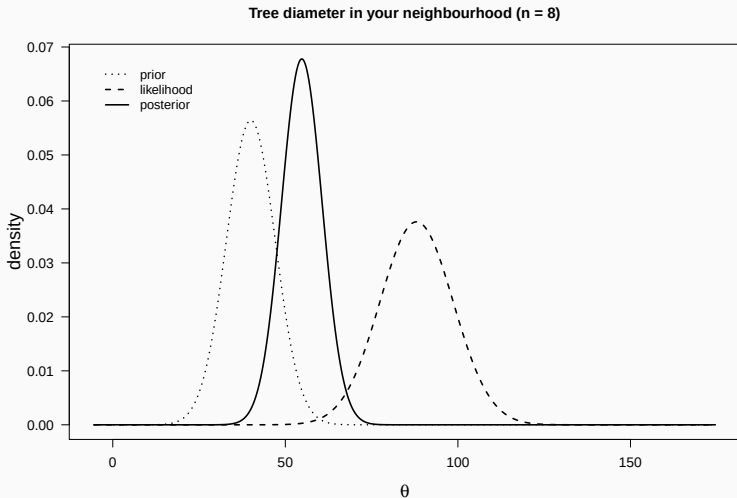
What is the likelihood?

$$L(\theta|x) = P(x|\theta)$$

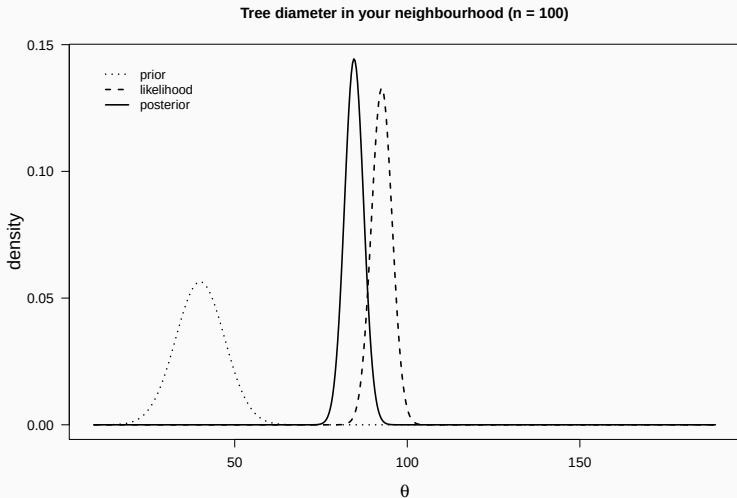


<https://seeing-theory.brown.edu/bayesian-inference/index.html>

Bayesian inference: prior and likelihood produce posterior



With increasing sample size, likelihood dominates prior



More apps to introduce Bayesian inference

- Wagenmaker's first lesson in Bayesian inference

More apps to introduce Bayesian inference

- Wagenmaker's first lesson in Bayesian inference
- Bayesian Demo

More apps to introduce Bayesian inference

- Wagenmaker's first lesson in Bayesian inference
- Bayesian Demo
- Bayesian inference for a population mean

More apps to introduce Bayesian inference

- Wagenmaker's first lesson in Bayesian inference
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- Own data

More apps to introduce Bayesian inference

- Wagenmaker's first lesson in Bayesian inference
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- Own data
- Bayesian t-test

- Integrate information (prior)

Bayesian statistics in practice

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- Prior regularises unlikely estimates from data

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- Large dataset -> prior effect diminishes

Bayesian statistics in practice

- Integrate information (prior)
- Prior regularises unlikely estimates from data
- Large dataset $\rightarrow$ prior effect diminishes
- Uncertainty / Propagate errors

The base rate fallacy

Do you have COVID?

- You test positive

Do you have COVID?

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- Test can detect positive COVID 90% of times (sensitivity)

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- Currently 1% people have COVID

Do you have COVID?

- You test positive
- Test can detect positive COVID 90% of times (sensitivity)
- Test correctly rejects COVID 80% of times (specificity)
- Currently 1% people have COVID
- What's the probability that you have COVID?

The probability
that you have COVID
is 0.04!

The probability that you have COVID is 0.04!



<https://seeing-theory.brown.edu/bayesian-inference/index.html>

The base rate fallacy

- [Matthews 2025](#)

The base rate fallacy

- [Matthews 2025](#)
- [Statistics done wrong](#)

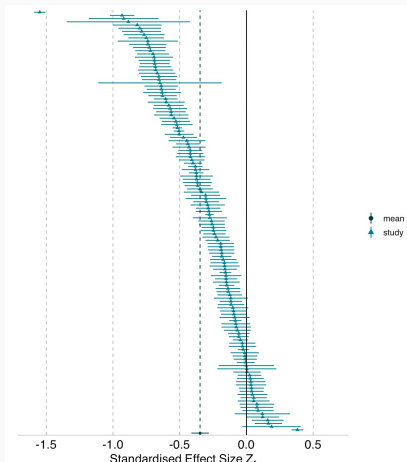
Uncertainty!

Same data, different teams reach different conclusions!

How does sibling competition affect nestling growth in blue tits?

Most teams found negative effect, but large variation in effect size.

Some teams no or positive effect



73 teams testing the same hypothesis with the same data

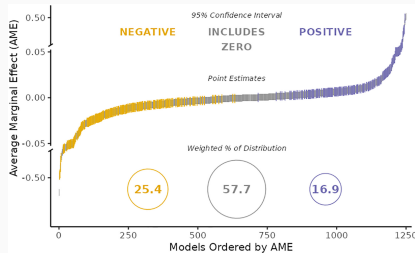
PNAS

RESEARCH ARTICLE

SOCIAL SCIENCES

OPEN ACCESS

Observing many researchers using the same data and hypothesis reveals a hidden universe of uncertainty



“This reveals a **universe of uncertainty** that remains hidden when considering a single study in isolation”

“These results call for greater **epistemic humility** and **clarity in reporting** scientific findings”

Experimental Design

How would you evaluate fertilizer effect?

Discuss with partner (5')



Experimental design principles

Replication

Replication!



Replication

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- Replication is key: we need several samples.
- How many? The more the better! See [Gelman & Carlin 2014](#).
- But there are trade-offs. See [Lakens' sample size justification](#)

The most important aspect of a statistical analysis is not what you do with the data, it's what data you use

The importance of sample size

- Many studies have **too low sample sizes**.

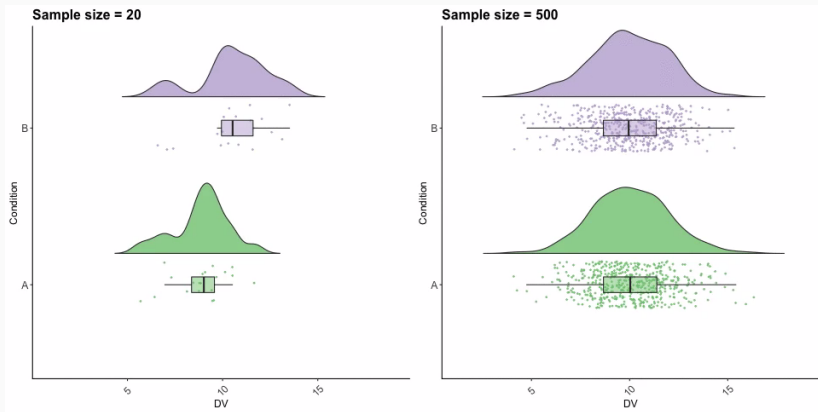
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The importance of sample size

- Many studies have **too low sample sizes**.
- Low sample sizes **miss subtle effects**, but also **prone to bias**.
- The **fallacy** of assuming that which does not kill statistical significance makes it stronger ([Loken & Gelman 2017](#)).

Low sample sizes are very sensitive to random noise



https://twitter.com/ajstewart_lang/status/1020038488278945797

Play yourself [here](#)

Low sample sizes may bias inferences about population



Source: statisticalgate.com

Low sample sizes may bias inferences

See *The evolution of correlations*

Stopping rules

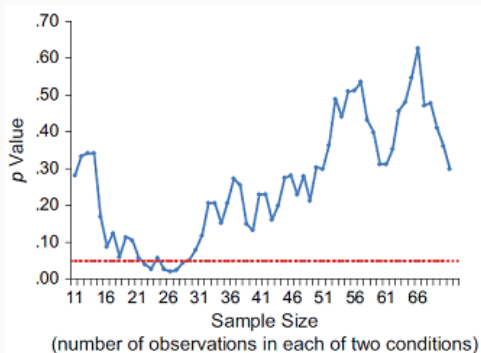


Fig. 2. Illustrative simulation of p values obtained by a researcher who continuously adds an observation to each of two conditions, conducting a t test after each addition. The dotted line highlights the conventional significance criterion of $p \leq .05$.

Sample size estimation

- Plan model/statistical analysis **before** data collection.

Sample size estimation

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- **Justify** your sample size (precision, minimal effect size...). See [Lakens' book](#).
- Plan to have at least **10-30 observations per predictor**.
- Complex models (w/ many predictors, interactions etc) require **high** sample sizes.

Sample size estimation

Calculating sample size for Gaussian (Normal) response model:

- expected mean: 30
- expected sd: 10
- 10 parameters (predictors)
- expected R-squared: 0.2

```
library(pmsampsize)
pmsampsize(type = "c", parameters = 10, intercept = 30, sd = 10, rsquared = 0.2)
```

NB: Assuming 0.05 acceptable difference in apparent & adjusted R-squared

NB: Assuming MMOE <= 1.1 in estimation of intercept & residual standard deviation

SPP - Subjects per Predictor Parameter

	Samp_size	Shrinkage	Parameter	Rsqr	SPP
Criteria 1	313	0.900	10	0.2	31.3
Criteria 2	161	0.827	10	0.2	16.1
Criteria 3	244	0.876	10	0.2	24.4
Criteria 4*	313	0.900	10	0.2	31.3
Final	313	0.900	10	0.2	31.3

Minimum sample size required for new model development based on user inputs = 313

* 95% CI for intercept = (29.01, 30.99), for sample size n = 313

Sample size estimation

Calculating sample size for binary response model:

- expected prevalence: 0.1
- 20 parameters (predictors)
- expected R-squared: 0.2

```
library(pmsampsize)
pmsampsize(type = "b", parameters = 20, prevalence = 0.1, nagrsquared = 0.2)
```

NB: Assuming 0.05 acceptable difference in apparent & adjusted R-squared

NB: Assuming 0.05 margin of error in estimation of intercept

NB: Events per Predictor Parameter (EPP) assumes prevalence = 0.1

	Samp_size	Shrinkage	Parameter	CS_Rsq	Max_Rsq	Nag_Rsq	EPP
Criteria 1	1774	0.900	20	0.096	0.478	0.201	8.87
Criteria 2	786	0.801	20	0.096	0.478	0.201	3.93
Criteria 3	139	0.900	20	0.096	0.478	0.201	0.70
Final	1774	0.900	20	0.096	0.478	0.201	8.87

Minimum sample size required for new model development based on user inputs = 1774,
with 178 events (assuming an outcome prevalence = 0.1) and an EPP = 8.87

Randomization

Randomization



Randomization

- Haphazard $\neq$ Random

Randomization

- Haphazard $\neq$ Random
- Stratify: randomize within groups (e.g. species, soil types)

Controls

Have controls

- Untreated individuals, plots... (assigned randomly, of course).

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- **Must differ only in treatment** (i.e. homogeneous environment).

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- Measure **before & after** treatment.

Have controls

- Untreated individuals, plots... (assigned randomly, of course).
- **Must differ only in treatment** (i.e. homogeneous environment).
- Measure **before & after** treatment.
- Consider **blind designs** to avoid observer bias.

Experimental design principles

1. Replication

Experimental design principles

1. Replication
2. Randomization

Experimental design principles

1. Replication
2. Randomization
3. Controls

To read more

- Ruxton & Colegrave: Experimental Design for the Life Sciences.
OUP

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OUP
- Lakens: Improving your statistical inferences

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- DeclareDesign

Inference from observational studies

News: Hamburgers increase risk of heart attack

- In a sample of 10,000 people, it was found that people eating >2 hamburgers a week had 20% higher probability of heart attack.

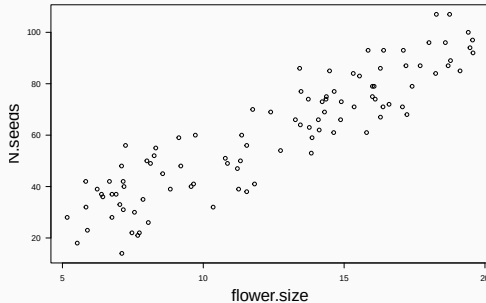
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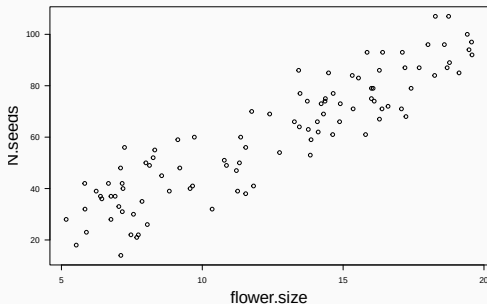
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- Do hamburgers increase heart attacks?
- <https://pollev.com/franciscorod726>

Bigger flowers increase reproductive success



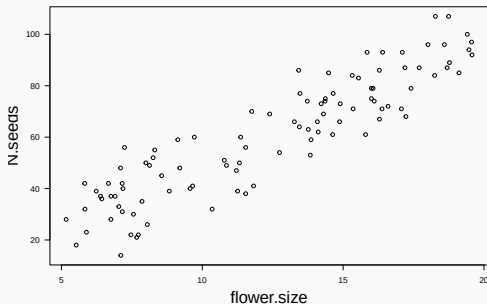
- We found that plants with big flowers produced 30% more seeds...

Bigger flowers increase reproductive success



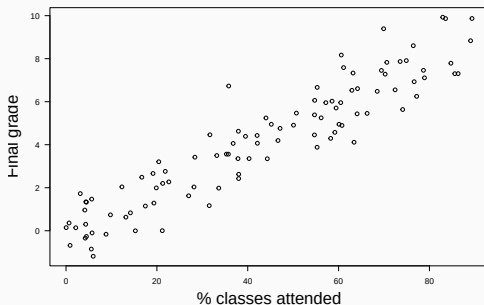
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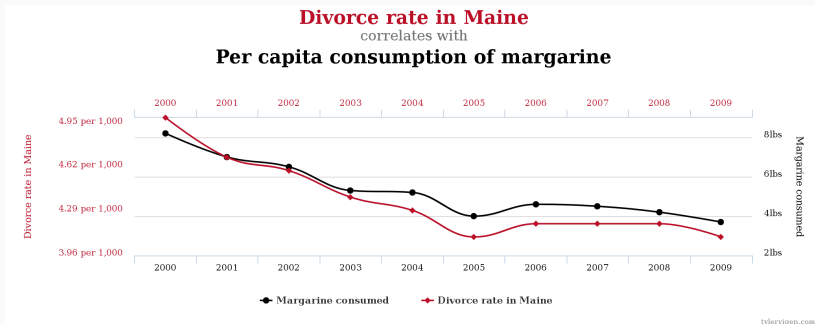
- We found that plants with big flowers produced 30% more seeds...
- Do big flowers increase reproductive success?
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Does coming to class improve students' performance?



- <https://pollev.com/franciscorod726>

Correlation vs Causation



<http://tylervigen.com/spurious-correlations>

Hypothesis testing

NHST concepts

Null and alternative hypotheses

- Tell me...

Null and alternative hypotheses

- Tell me...
- **Null hypothesis:** there is no difference between groups.

Null and alternative hypotheses

- Tell me...
- **Null hypothesis:** there is no difference between groups.
- **Alternative hypothesis:** groups are different.

Are there any differences? A non-sensical question in ecology

Alejandro Martínez-Abraín

IMEDEA (CSIC-UIB), C/Miquel Marquès 21, 07190 Esporles, Majorca, Spain

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ABSTRACT

One of the main questions that ecologists pose in their investigations includes the analysis of differences in some trait between two or more populations. I argue here that asking whether there are differences or not between populations is biologically irrelevant, since **no two living things are ever equal**. On the contrary **the appropriate question to pose is how large differences are between populations**. That is, **we urge a shift in interest from statistical significance to biological relevance** for proper knowledge accumulation. I empha-

What is the p-value?

- The probability that the observed data were produced by chance

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 - Null hypothesis is true

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- What assumptions?
 - Null hypothesis is true
 - No uncontrolled sources of bias (measurement or programming error, p-hacking, etc)

A very small p-value does NOT mean H_0 is automatically false

- A famous experiment found neutrinos faster than light

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- p-value $< 10^{-7}$ $\rightarrow$ reject null hypothesis of equal speed

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- A famous experiment found neutrinos faster than light
- $p\text{-value} < 10^{-7} \rightarrow$ reject null hypothesis of equal speed
- In reality, measurement error (loose cable)

Testing a new fertiliser: Does it work?

10 treatment, 10 control plots

Crop biomass (treatment): $10\text{kg} \pm 2\text{kg}$

Crop biomass (control): $6\text{kg} \pm 2\text{kg}$

Welch Two Sample t-test

```
data:  tr and ct
```

```
t = 4.1858, df = 17.872, p-value = 0.0005631
```

```
alternative hypothesis: true difference in means is not equal to 0
```

```
95 percent confidence interval:
```

```
 1.857902 5.606112
```

```
sample estimates:
```

```
mean of x mean of y
```

```
10.149251  6.417244
```

Testing a new fertiliser: Does it work?

10 treatment, 10 control plots

Crop biomass (treatment): $10\text{kg} \pm 2\text{kg}$

Crop biomass (control): $9\text{kg} \pm 2\text{kg}$

Welch Two Sample t-test

```
data:  tr and ct
```

```
t = 0.82102, df = 17.872, p-value = 0.4225
```

```
alternative hypothesis: true difference in means is not equal to 0
```

```
95 percent confidence interval:
```

```
-1.142098  2.606112
```

```
sample estimates:
```

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```

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How to interpret P-values

- **Low P-value:** data unlikely if every model assumption (including H_0) were true.

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- See [Greenland et al 2016](#)

If $p\text{-value} > 0.05$

- the null hypothesis is false, i.e. the alternative hypothesis must be true

<https://pollev.com/franciscorod726>

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- there is no difference between groups

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Common practice:

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<https://doi.org/10.1038/d41586-019-00857-9>

S (Surprise) values as alternative to p-values

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S (Surprise) values as alternative to p-values

- A p-value of 0.05 is as surprising as getting 4.3 heads in a row when flipping a coin.
- A p-value of 0.01 is as surprising as getting 6.6 heads in a row when flipping a coin.
- A p-value of 0.001 is as surprising as getting 10 heads in a row when flipping a coin.

Are these two groups different?

```
t.test(group.A, group.B)
```

Welch Two Sample t-test

data: group.A and group.B

t = -5.0271, df = 12.464, p-value = 0.0002632

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:

-24.91029 -9.88971

sample estimates:

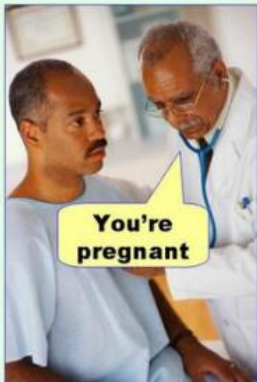
mean of x mean of y

162.8 180.2

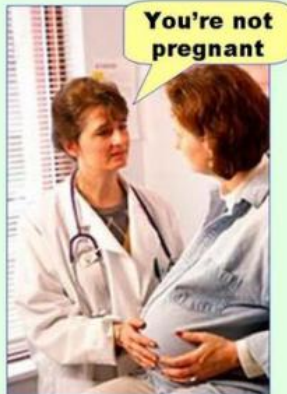
<https://pollev.com/franciscorod726>

Rejecting hypotheses: two types of error

Type I error
(false positive)



Type II error
(false negative)



Rejecting hypotheses: two types of error

Statistics: Hypothesis Test	Null Hypothesis is True	Null Hypothesis is False
	Type I Error	Correct
Reject Null Hypothesis		
Fail to Reject Null Hypothesis	Correct	Type II Error

POWER: Probability of detecting true difference (rejecting H_0 when it's false).

Is this coin biased?

```
[1] 0 0 0 1 1 1 1 0 1 1
```

1-sample proportions test with continuity correction

data: sum(coin) out of ntrials, null probability 0.5

X-squared = 0.1, df = 1, p-value = 0.7518

alternative hypothesis: true p is not equal to 0.5

95 percent confidence interval:

0.2736697 0.8630694

sample estimates:

p

0.6

<https://pollev.com/franciscorod726>

<http://rpsychologist.com/d3/NHST/>

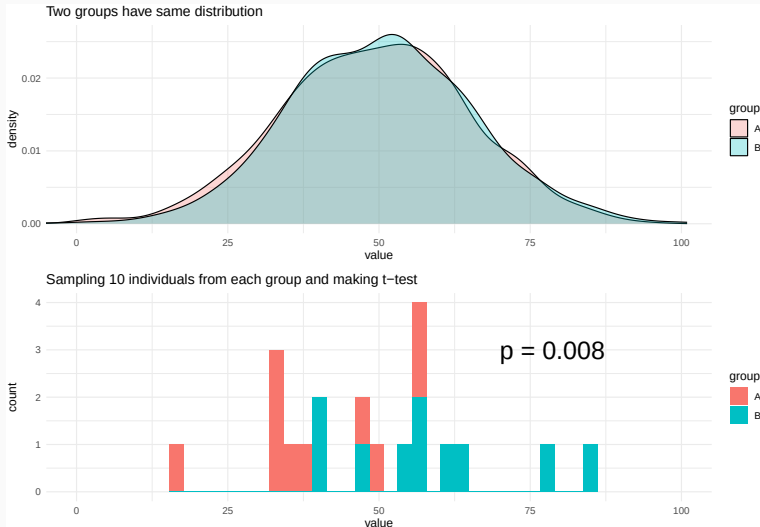
[https://lakens.github.io/statistical_inferences/
01-pvalue.html](https://lakens.github.io/statistical_inferences/01-pvalue.html)

NHST and p-values: common pitfalls

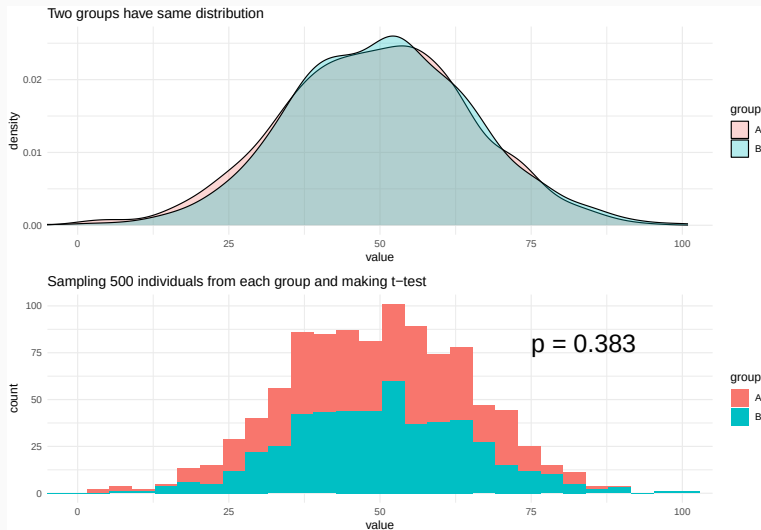
A significant p-value
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A significant p-value does not mean we found a true difference

Particularly with low sample sizes



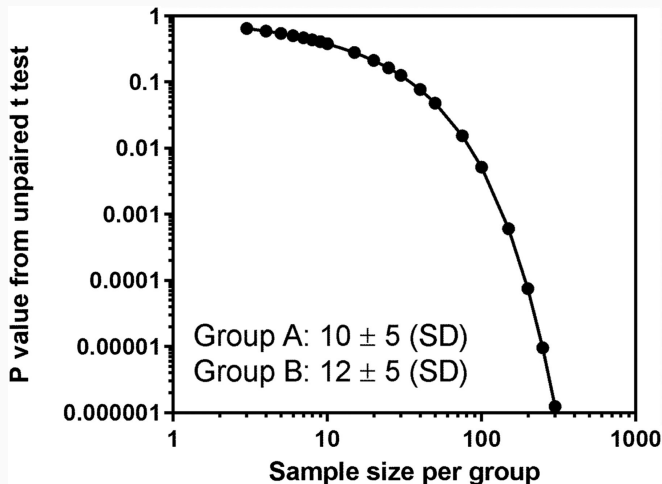
If sample size was larger...



With low sample size (power),
significant p-values
are most likely overestimates

Loken & Gelman 2014, Vasisth et al. 2018

P-value depends on sample size



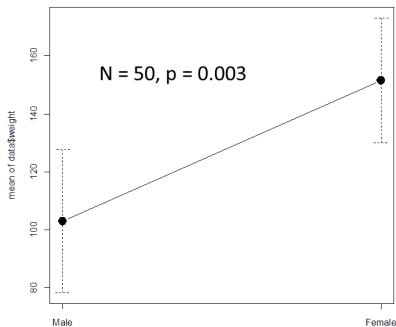
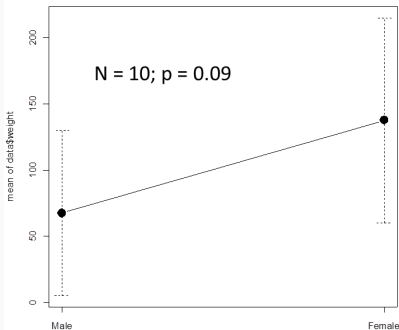
<https://doi.org/10.1002/prp2.93>

Try yourself [here!](#)

P-value depends on sample size

Same real difference is detected as significant or not depending on sample size

Real difference = 40 g



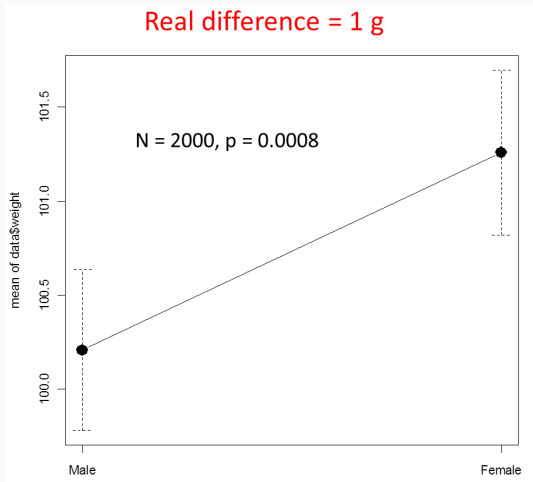
Statistically significant

!=

biologically important

Statistically significant != biologically important

With big sample size, we can find **highly significant** but **biologically unimportant** differences.



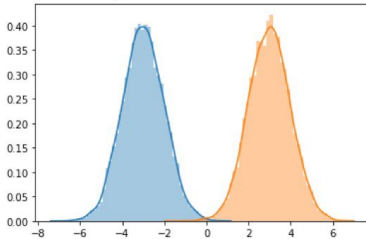
Statistically significant != biologically important



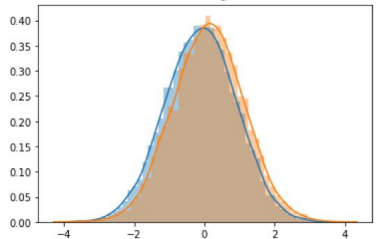
zara weinberg
@weinberz

friendly reminder about $p < 0.0001$:

What you think it means:

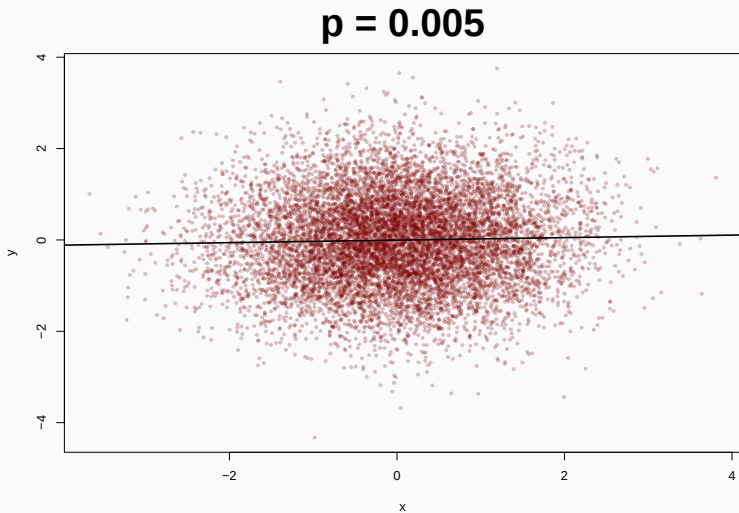


What it actually means:



<https://twitter.com/weinberz/status/1422405165236178947?s=20>

Statistically significant != biologically important



Statistically significant != biologically important

- Statistically significant = unlikely to be zero

Statistically significant != biologically important

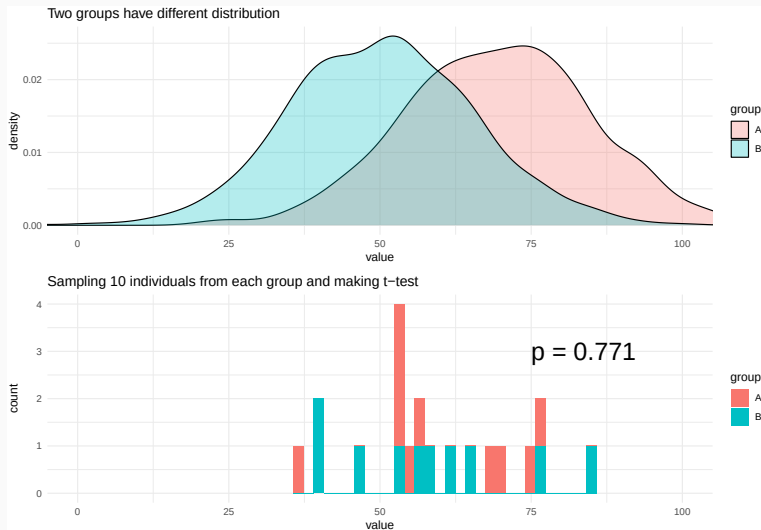
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- Good read: *significantly misleading*

Statistically significant != biologically important

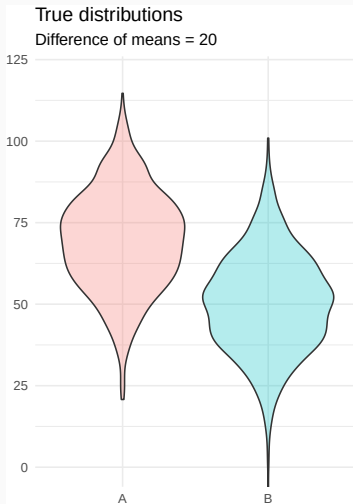
- Statistically significant = unlikely to be zero
- Good read: *significantly misleading*
- Beyond significant/not significant, look at **effect sizes and their uncertainty**.

‘Not significant’
does NOT mean
‘there is no effect’

'Not significant' does NOT mean 'they are equal'



'Not significant' does NOT mean 'there is no effect'



Failure to reject H_0 $\neq$ H_0 is true

Absence of evidence $\neq$ Evidence of absence

- “We were **unable to find evidence** against the hypothesis that $A = B$ with the current sample size” ([Harrell](#))

- “We were **unable to find evidence** against the hypothesis that $A = B$ **with the current sample size**” ([Harrell](#))
- “Differences between groups were **not statistically clear**” ([Dushoff et al](#))

Is it safe to allow right turn with red lights?

- Right turn not allowed: 308 accidents



<https://www.statisticsonewrong.com/power.html#the-wrong-turn-on-red>

Is it safe to allow right turn with red lights?

- Right turn not allowed: 308 accidents
- Right turn allowed: 337 accidents



<https://www.statisticsonewrong.com/power.html#the-wrong-turn-on-red>

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- Right turn allowed: 337 accidents
- No *significant* difference, hence safe



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Is it safe to allow right turn with red lights?

- Right turn not allowed: 308 accidents
- Right turn allowed: 337 accidents
- No *significant* difference, hence safe
- Failure to reject H_0 does NOT mean H_0 is true!
- Misinterpretation of underpowered study cost lives



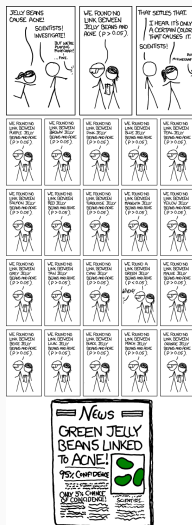
<https://www.statisticsonewrong.com/power.html#the-wrong-turn-on-red>

The Difference Between “Significant” and “Not Significant” is not Itself Statistically Significant

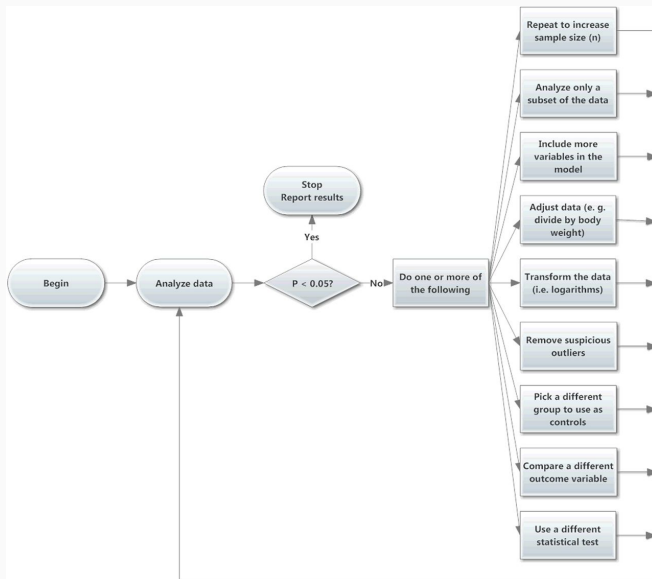
Andrew GELMAN and Hal STERN

<https://dx.doi.org/10.1198/000313006X152649>

Multiple hypothesis testing



How to make your results significant: *p*-hacking



How to make your results significant: *p-hacking*

1. Test multiple variables, then report the ones that are significant.

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 2. Artificially choose when to end your experiment.
 3. Add covariates until effects are significant.
 4. Test different conditions (e.g. different levels of a factor) and report the ones you like.
- To read more: [Simmons et al 2011](#)

p-hacking: try it yourself

<https://www.shinyapps.org/apps/p-hacker/>

<https://shiny.psy.lmu.de/felix/ShinyPHack/>

How to make your results significant: *p-hacking*

<https://www.youtube.com/watch?v=ZaNtz76dNSI>

ASA statement on p-values

- P-values do not measure the **probability of hypothesis** being true, or the probability that the data were produced by **random chance** alone.

<https://doi.org/10.1080/00031305.2016.1154108>

See also https://lakens.github.io/statistical_inferences/01-pvalue.html

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- By itself, a p-value does NOT provide a good **measure of evidence** regarding a model or hypothesis.

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Good practice

Eur J Epidemiol (2016) 31:337–350

DOI 10.1007/s10654-016-0149-3



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ESSAY

Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations

Sander Greenland¹ · Stephen J. Senn² · Kenneth J. Rothman³ · John B. Carlin⁴ · Charles Poole⁵ · Steven N. Goodman⁶ · Douglas G. Altman⁷

<https://doi.org/10.1007/s10654-016-0149-3>

esa

ECOSPHERE

Applied statistics in ecology: common pitfalls and simple solutions

E. ASHLEY STEEL,^{1,†} MAUREEN C. KENNEDY,² PATRICK G. CUNNINGHAM,³ AND JOHN S. STANOVICK⁴

<https://doi.org/10.1890/ES13-00160.1>

Also <http://www.statisticsonewrong.com/>



Twenty tips for interpreting scientific claims

Aim for estimation of effects and their uncertainty (SE, CI...)

General Article

The New Statistics: Why and How

Geoff Cumming
La Trobe University

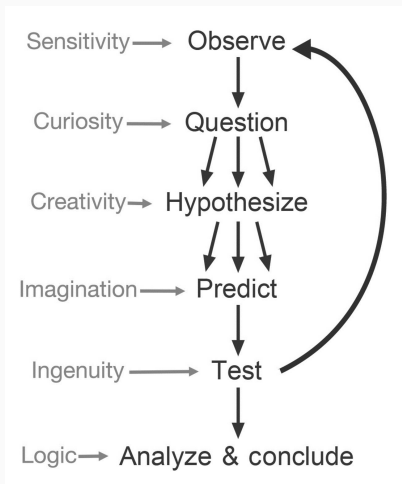


Psychological Science
2014, Vol. 25(1) 7–29
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DOI: 10.1177/0956797613504966
pss.sagepub.com



<https://dx.doi.org/10.1177/0956797613504966>

Instead of falsifying null model, compare meaningful models



<https://doi.org/10.1242/jeb.104976>

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- Beyond Power Calculations: Assessing Type S (Sign) and Type M (Magnitude) Errors



Source code and materials:

<https://github.com/Pakillo/stats-intro>